

MEHMET FIRAT

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Technical Skills

CAD & Analysis: SolidWorks, Onshape, finite element analysis (FEA), design for manufacturing (DFM)

Manufacturing: CNC machining, waterjet cutting, laser cutting, welding, 3D printing, drill press, band saw, angle grinder

Programming & Simulation: MATLAB, Simulink, Python, Arduino/C++, Java

Work Experience

Cibo Vita

Jun 2026 — Aug 2026

Manufacturing Engineering / Operations Intern

Totowa, NJ

- Designed a CAD cutter-scraper extension; selected material and mounting approach and supported installation/testing, *eliminating roughly 2-minute cleaning stops occurring hourly.*
- Designed a cleaning cart for 120 MULTIPOND parts with loading, rinse access, drainage, durability, and weight constraints.
- Iterated 3D-printed holders through about 30 tests per hopper type to improve fit, retention, and ease of loading.
- Investigated low output on about 5 production machines; identified hopper-discharge clogging and proposed improvements.

Boston Education & Counseling

Oct 2025 — Present

Robotics Mentor

- Mentor students through robot builds and troubleshoot mechanical design, wiring, fabrication, and assembly issues.

Education

Olin College of Engineering, Needham, MA

Expected May 2028

B.S. in Mechanical Engineering

GPA: 3.92/4.00

Relevant Coursework: Mechanical Design (in progress); Mechanics of Solids & Structures; Thermal-Fluid Systems Analysis; Engineering Systems Analysis: Dynamics; Sensors, Instrumentation & Measurement

Project Experience

HydroFleet Drone-Deployed Soil-Moisture Probe | *Olin College Robotics Lab*

Sep 2024 — Present

Mechanical Design Team Member; Mechanical Design Co-Lead since Fall 2026

- Co-developed two drone-mounted soil-probe prototypes, *achieving 6-inch insertion at 650 g within a 700 g payload limit.*
- Tested anchor geometry to resist drill-induced uplift and improve stability during soil insertion.
- Team received the 2026 Farm Robotics Challenge Excellence in Drone Applications Award (Division I).

Masterpiece Automated Chessboard | *Olin Principles of Integrated Engineering*

2025

Team mechanical design project

- Designed board mounts and co-designed belt tensioners, magnet carriage, and side panels for a 35 × 28 in. CoreXY system.
- Integrated laser-cut and 3D-printed components into a concealed gantry for magnetic chess-piece movement.

Rocky Self-Balancing Robot | *Olin Engineering Systems Analysis*

2026

Controls and robotics team project

- Identified motor and pendulum dynamics from experimental data and developed MATLAB/Simulink models of a two-wheel inverted-pendulum robot.
- Developed and implemented a five-pole feedback controller in Arduino, enabling the robot to recover angle, velocity, and position after disturbances.

Formula Dashboard & Electronics Enclosure | *Olin Formula Team*

Sep 2024 — May 2026

Cockpit Team Member; Drivetrain Mentor (2025–26)

- Designed and fabricated a SolidWorks dashboard/electronics enclosure using FEA, *reducing mass by about 20%.*
- Added an LED sun guard and removable lid and refined mounts and connector packaging through cross-team design reviews.